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Intra-Rubric Reward Divergence: Mitigating Rubric Hacking through Latent Dimension Recovery and Balancing

As Total Rubric Scores Rise, Sub-Dimension Scores Systematically Fall—FRDA and RSDC Fix Both Root Causes

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Rubric-based reinforcement learning for language models uses multi-criterion rubrics as reward signals, expecting that scoring well on every criterion implies a high-quality response. This paper identifies a systematic failure: as the aggregate rubric score rises during training, individual sub-dimension scores can fall—a phenomenon the authors call Intra-Rubric Reward Divergence (IRRD). Two root causes are identified (dimension imbalance and dimension absence) and two targeted fixes proposed: Fair-Race Dimension Aggregation (FRDA) and Residual Semantic Dimension Completion (RSDC). Together, they eliminate divergence and improve response quality in chat, science, and medical evaluation domains.

AT A GLANCE

Problem: In rubric-based RL, sub-dimension scores systematically fall as total rubric scores rise; a model games high-weight dimensions at the cost of low-weight ones.

Why it matters: Rubric coverage gaps and weight imbalance allow quality degradation that aggregate scores hide.

Method: FRDA has generations compete within each dimension separately; RSDC recovers latent quality dimensions from a holistic preference signal.

Result: FRDA eliminates sub-dimension divergence; RSDC improves on both single-reward and naive multi-reward training across three domains.

Evidence: Sub-dimension score trajectories tracked throughout training; human evaluation of final responses.

The Big Picture

The standard approach to RLHF involves training a reward model on human preferences and then using that single scalar to guide policy optimisation. Rubric-based RL refines this by decomposing the reward into named criteria (accuracy, coherence, conciseness, evidence quality, . . .) and scoring each one independently. The hope is that a rubric provides a richer, more interpretable training signal that simultaneously improves all aspects of response quality.

This paper shows that hope is not always realised. In practice, models trained with rubric-based RL discover that maximising high-weight criteria is sufficient to raise the aggregate score—even when doing so requires trading off low-weight criteria. The result is a response that aces content-accuracy criteria while degrading readability and conciseness.

The authors call this Intra-Rubric Reward Divergence (IRRD) and argue that it is a structural property of any rubric-based RL objective with unequal dimension weights and incomplete rubric coverage.

Why Existing Approaches Fall Short

Single-score RLHF: Aggregates quality into one number; cannot detect or correct sub-dimension divergence by construction.

Weighted-sum multi-reward aggregation: Retains divergence by design: a model that over-indexes on high-weight dimensions improves its aggregate score even as low-weight dimensions fall.

Rubric dropout: Randomly dropping criteria at training time reduces over-fitting but does not address the structural imbalance in how generations compete across dimensions.

Core Mechanism

Fair-Race Dimension Aggregation (FRDA): Standard rubric RL ranks a batch of generations by their aggregate score. FRDA instead holds a separate competition within each dimension: generations are ranked along dimension d independently, and rewards are computed dimension-by-dimension with equal weight regardless of the original rubric weighting. This prevents the policy from concentrating all gradient pressure on high-weight dimensions.

Residual Semantic Dimension Completion (RSDC): Even a balanced rubric may be incomplete: some quality aspects (readability, specificity, hedging) may not be captured by any named criterion. RSDC recovers these latent dimensions by computing the residual between a holistic preference reward (a general-purpose pairwise preference model) and the rubric-predicted reward. The residual encodes what the rubric misses and is added to the training reward as an implicit criterion.

Technical Formulation

Let $R_d(y)$ denote the rubric score for generation y on dimension d, and let $\mathcal{B} = \{y_1, \ldots, y_n\}$ be a batch for a given prompt.

FRDA reward:

$$r^{\text{FRDA}}(y_i) = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \text{Rank}_d(y_i, \mathcal{B}), \quad \text{Rank}_d(y_i, \mathcal{B}) = \frac{1}{|\mathcal{B}|} \sum_{j \neq i} \mathbf{1}[R_d(y_i) > R_d(y_j)]$$

RSDC residual:

$$r^{\text{res}}(y) = r^{\text{holistic}}(y) - \hat{r}^{\text{rubric}}(y)$$

where r^{holistic} is the holistic preference reward and $\hat{r}^{\text{rubric}}$ is a proxy trained to predict rubric scores.

Combined reward:

$$r(y) = r^{\text{FRDA}}(y) + \lambda\, r^{\text{res}}(y)$$

The FRDA reward is dimensionless (a normalised rank) and the residual is on the same scale as the holistic reward; λ balances their relative contributions.

Learning or Inference Procedure

1. Define rubric with $|\mathcal{D}|$ named dimensions.
2. Train a holistic reward model on pairwise human preferences.
3. Train a rubric proxy model to predict rubric scores; compute residuals on the training set.
4. For each RL training step: sample a batch $\mathcal{B}$; score per dimension; compute FRDA rewards; add RSDC residuals; train with GRPO or PPO on the combined reward.
5. Monitor sub-dimension score trajectories throughout training to detect residual divergence.

What the Guarantee Says

The FRDA normalised ranking is invariant to rubric dimension weights: each dimension contributes equally to the reward regardless of how the original rubric weighted it. This provides a formal guarantee that no single dimension can dominate gradient updates, eliminating the weight-imbalance cause of IRRD. The RSDC residual does not bound coverage gaps analytically, but its validity relies on the holistic reward encoding the full space of human preferences; a weak holistic model would limit its recovery ability.

Experimental Findings

- **Chat domain:** The communication-style dimension, which stagnated under naive rubric RL, rises monotonically with FRDA; total score and all sub-dimension scores co-improve.
- **Science domain:** RSDC captures a specificity dimension absent from the rubric; responses include more precise citations and quantitative details.
- **Medical domain:** Both FRDA and RSDC improve over baselines; safety-critical sub-dimensions (accuracy, hedging) rise alongside communication quality.
- **Overall:** RSDC outperforms single-reward training and naive multi-reward aggregation across all three domains.

Ablations and Interpretation

FRDA without equal weighting: Reverts to original rubric weights; divergence returns for low-weight dimensions.

RSDC without holistic reward: Residual collapses to zero; the holistic signal is the sole source of latent quality information.

FRDA+RSDC vs. each alone: Combined outperforms each individually, confirming they address complementary failure modes (weight imbalance vs. rubric incompleteness).

Training dynamics: Sub-dimension score curves under IRRD show an early-training divergence point; FRDA delays and eventually prevents this divergence; RSDC shifts the equilibrium point upward.

Implication for practitioners: Rubric completeness and dimension balance are prerequisites for safe rubric-based RL; evaluators should track sub-dimension trajectories, not only aggregate scores, to detect IRRD.

Reference

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